

Ontological innovation: This project is conceived of as a pilot study in applying linguistic ontologies to machine learning/NLP. By testing a feature-based ontology on PLCs — a notoriously complex semantic domain — the project demonstrates how ontological feature schemas can mediate between linguistic theory and computational modeling. This opens pathways for future research applying similar methods to other phenomena that have thus far proven challenging for existing LLMs and NLP systems, such as distinguishing embedded versus nominal clauses, robustly modeling linguistic nuance in areas like TAM and linguistic ambiguity, and perhaps most importantly improving accessibility of digital resources to low-resource, underdescribed, and/or typologically diverse languages. Ontology-driven feature schemas and modeling pipelines tested on PLCs could serve as a prototype for tackling these problems and contributing to a broader program whereby linguistic ontologies facilitate cross-disciplinary advances.

Key/Core Research Questions

Core points to get across:

- ☐ **Core question:** To what extent can a feature-based semantic ontology, developed from aligned parallel corpus data, model cross-linguistic variation in PLCs?
- ☐ **Core question:** Does the application of such a schema reveal typologically consistent contrasts in meaning (e.g. experientiality, current relevance, etc.) across languages that differ morphosyntactically and genealogically, offering a scalable and interpretable framework for typological comparison, semantic annotation, and the representation of temporally complex meanings in both research and applied settings?
- ☐ **Concomitant questions:**
 - ☐ Is the feature-based schema internally coherent and usable?
 - ☐ Can these features be defined clearly?
 - ☐ Are they recognizable across languages? (or statistically or typologically aligned subgroups of languages)
 - ☐ Can they be reliably annotated?
 - ☐ How do individual languages express these features?
 - ☐ What features do English PLCs encode versus other languages' PLCs?
 - ☐ Are there recurring patterns? Shared strategies?
 - ☐ Can we group languages or constructions by their feature “profiles”?

Background & Lit Review

Core points to get across:

- ☐ Formal / semantic approaches (Klein, Iatridou, Pancheva, etc., interface debates [Aktionsart, TAM, universals vs existentials, etc.]).
- ☐ Typological surveys (Dahl, Plungian, etc.).
- ☐ Computational / corpus approaches (Chen, van der Klis, etc.).
- ☐ Gap: no integrated, ontology-driven method that bridges linguistics + computation.

Formal/Semantic Approaches

Definitely discuss:

- | | |
|--|--|
| ☐ Alexiadou, Rathert, & von Stechow (2003) | ☐ Pancheva (2004) |
| ☐ Anagnostopoulou (1998) | ☐ Pancheva & von Stechow (2004) |
| ☐ Grønn & von Stechow (2021) | ☐ Portner (2003) |
| ☐ Iatridou et al. (2001) | ☐ Portner (2011) |
| ☐ Klein (1992) | ☐ Ritz (2012) |
| ☐ Löbner (2002, “Is the German Perfekt a perfect perfect?”) | ☐ Rothstein (2006) |
| ☐ McCawley (1981) | ☐ Schaden (2009) |
| ☐ McCoard (1976) | ☐ Stowell (2007) |
| ☐ Mittwoch (2008) | ☐ Stowell (2008) |
| ☐ Nishiyama Keonig (2010) | ☐ Zhao (2022a, 2022b) |
| ☐ Pancheva (2003) | |

Maybe discuss:

- | | |
|---|--|
| ☐ Abusch & Rooth (1990) | ☐ Katz (2011) |
| ☐ Carrasco (2015 “Perfect states”) | ☐ Kiparsky (2002) |
| ☐ Declerck (2006, “The absolute use of the present perfect”) | ☐ Kratzer (2004) |
| ☐ Franco & Tagliamonte (2022) | ☐ Lindstedt (2000) |
| ☐ Iatridou (2017) | ☐ Matthewson et al. (“Inchoativity...”) |
| ☐ Izvorski (1997, “The present perfect as an epistemic modal”) | ☐ Paslawska & von Stechow (2003, “Perfect readings in Russian”) |
| ☐ Kamp et al. (2015) | ☐ Werner (2013, “Temporal adverbials and the present perfect/past tense alternation”) |

Typology and Typological Surveys

- ☐ Bertrand et al. (2022)
- ☐ Bowler & Ozkan (2017)
- ☐ Dahl (2000, the PFQ)
- ☐ Dahl & Velupillai (2013)
- ☐ de Swart (2016)
- ☐ Haspelmath:
 - ☐ 2010a: “Comp concepts and desc cats in crossling studies”
 - ☐ 2010b: “The interplay btwn comp concepts and descr cats”
 - ☐ 2016: “The challenge of making lang desc and comp mutually beneficial”
 - ☐ 2021: “Explaining gram coding asym form-freq corr and predictability”
 - ☐ 2023: “Word-class universals and lang-particular analysis”
- ☐ Matthewson et al. (2014, “Miss Smith’s Bad Day”)
- ☐ Plungian (2011)

Computational/Corpus Approaches

- ☐ Chen et al. (2021)
- ☐ LeBruyn, van der Klis, & de Swart (2019, “The perfect in dialog”)
- ☐ van der Klis et al. (2017)
- ☐ van der Klis et al. (2021)
- ☐ van der Klis, LeBruyn, & de Swart (2021, “The present perfect: an unstable category in an overall stable tense-aspect grammar of the western European languages”)

Gap

Where does this project fit in? What issues does it seek to resolve? What are its contributions to the literature?

IV. Pilot Experiment

This section presents the pilot experiment which has been completed, and which informs and shapes the following work of the project.

Methodology

The data used in the pilot experiment are comprised of 300 tokens in three languages; the tokens are all pulled from New Testament Bibles (NTBs), and the languages in the experiment are English, German, and Chinese. Each token represents a distinct instance of a construction of interest (i.e. each unique occurrence of a PLC); these tokens were identified in a tokenized NTB using a simple Python script which locates, extracts, and compiles all instances of auxiliary *have* in co-occurrence with a verbal noun (as defined by spaCy; these instances are almost exclusively past passive participles) in the English NTB. The output of this script is a collection of verses, indexed by verse number, which contain one or more examples of a PLC. Next, the verse numbers are aligned with identically indexed verses in NTBs in the additional language of the study¹.

Once the data are extracted and aligned, the next step is annotation; note that here *annotation* refers to the process of examining each individual PLC in its contexts and coding it for semantic/ontological features and identifying which construction is used in each language. The primary disadvantage of this process, at least at this stage, is that as an L1 speaker of only one language (English), the experimenter is limited in their ability to identify semantic nuance in the non-English constructions, such that true equivalence (whether the aligned text encodes perfect-like semantics) and any additional semantic features (such as grammatically or pragmatically encoded evidentiality) are left to the annotator’s best interpretations. In subsequent studies, this will be addressed by recruiting undergraduate RA native speakers to conduct the non-English coded work; these RAs will work in tandem with the author using a clear and concise annotation manual and performing checks via inter-annotator agreement (IAA).

¹ The obvious limitation of this methodology is that it only identifies PLCs that encoded as *have*+participle perfects in English; subsequent experiments will use reverse lookup techniques to identify constructions of interest in each of the other languages as well.